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Activity-3

1. Scenario

Hospital Patient Admission Analysis

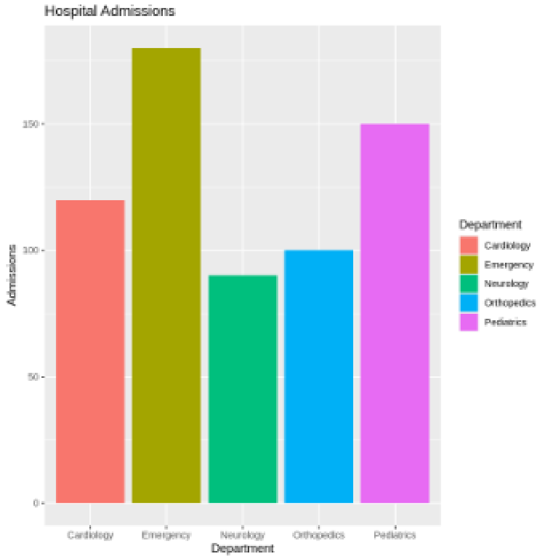
Student Task

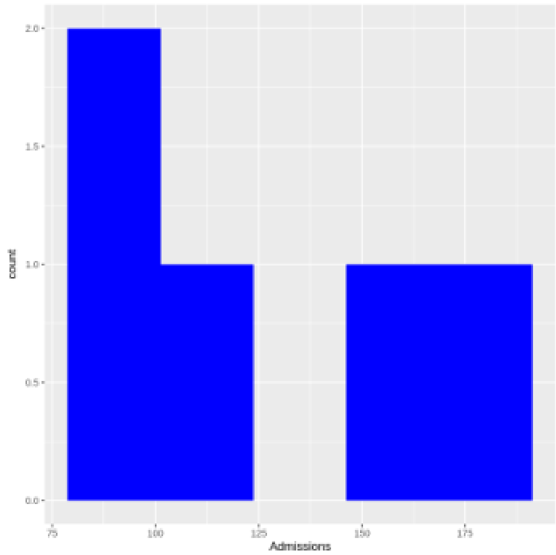
A hospital has collected data on patient admissions across multiple departments (Cardiology, Neurology, Pediatrics, Orthopedics, Emergency) for the past year. Different visualizations (bar chart, stacked bar chart, treemap, box plot, histogram) are available. Evaluate which visualization most effectively communicates department workload, admission variability, and patient distribution. Justify your selection with evidence.

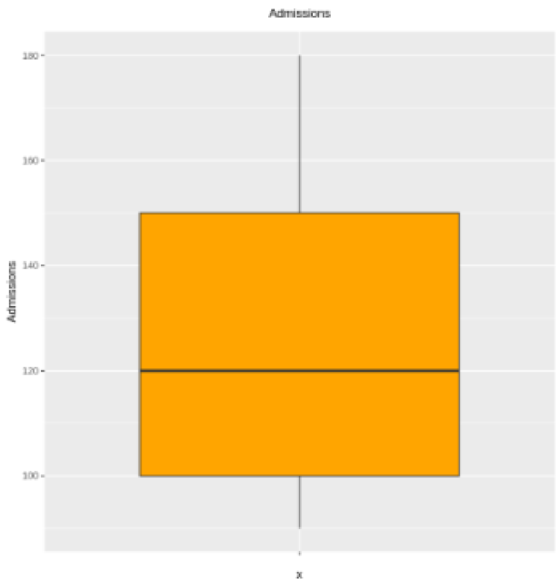
Code

```
library(ggplot2)
data <- data.frame(
Department=c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency"),
Admissions=c(120,90,150,100,180)
)
# Bar Chart
ggplot(data,aes(Department,Admissions,fill=Department))+
geom_bar(stat="identity")+
ggtitle("Hospital Admissions")
# Histogram
ggplot(data,aes(Admissions))+
geom_histogram(fill="blue",bins=5)
# Box Plot
ggplot(data,aes(x="",y=Admissions))+
geom_boxplot(fill="orange")
```

...







2. Scenario

E-Commerce Customer Spending Behaviour

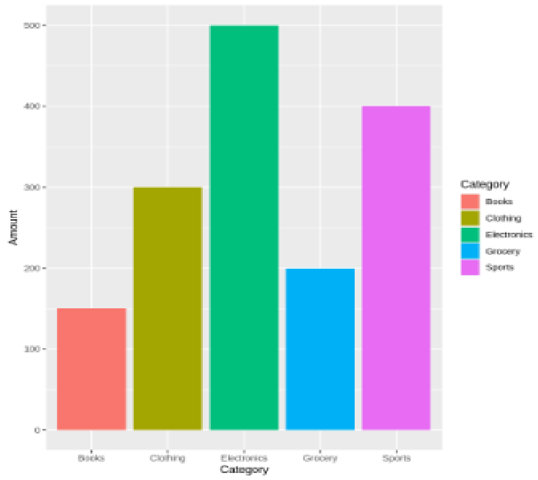
Student Task

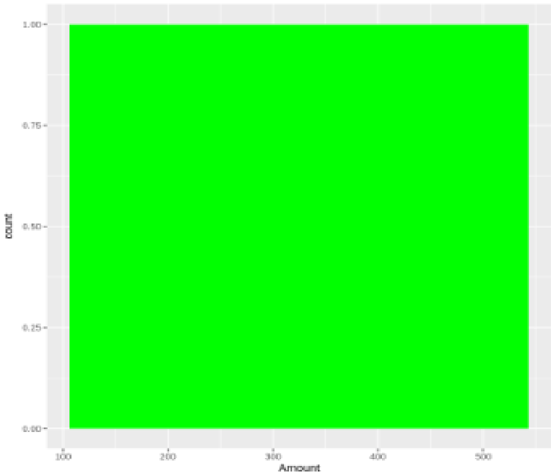
An online retailer has customer purchase amounts, order frequency, and product categories. Create multiple visualizations (histogram, violin plot, box plot, density plot, bar chart). Evaluate which visualization best identifies high-value customers, spending variability, and purchasing trends. Recommend the most appropriate visualization for business executives and explain why.

Code:

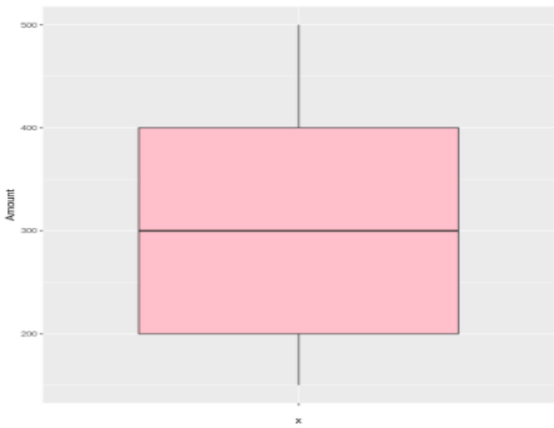
```
library(ggplot2)
data <- data.frame(
Category=c("Electronics","Clothing","Grocery","Books","Sports"),
Amount=c(500,300,200,150,400)
)
# Bar Chart
ggplot(data,aes(Category,Amount,fill=Category))+
geom_bar(stat="identity")
# Histogram
ggplot(data,aes(Amount))+
geom_histogram(fill="green",bins=5)
# Box Plot
ggplot(data,aes(x="",y=Amount))+
geom_boxplot(fill="pink")
# Density Plot
ggplot(data,aes(Amount))+
geom_density(fill="lightblue")
```

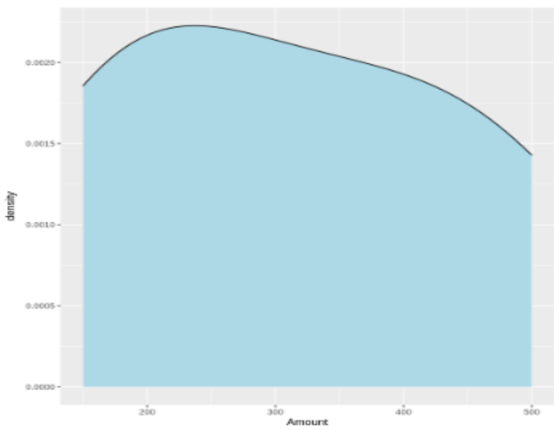
...


Amount
Category
Books
Clothing
Electronics
Grocery
Sports
0
100
200
300
400
500


count
Amount
0.00
0.25
0.50
0.75
1.00
100
200
300
400
500

...


Amount
x
200
300
400
500


density
Amount
0.0000
0.0005
0.0010
0.0015
0.0020
200
300
400
500

3. Scenario

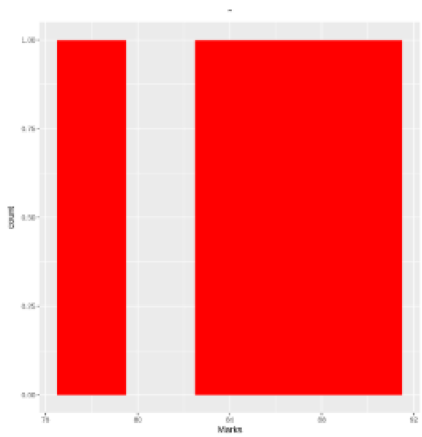
University Examination Performance Dashboard

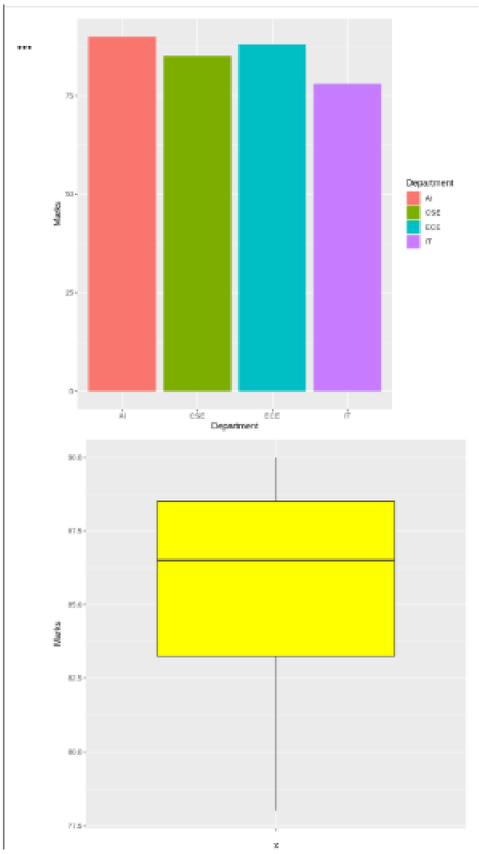
Student Task

A university wants to compare student marks across departments and identify score distributions, outliers, and top-performing classes. Generate different visualizations and evaluate which charts provide the clearest insights for academic decision-making. Recommend improvements to the visualization dashboard.

Code:

```
library(ggplot2)
data <- data.frame(
Department=c("CSE","AI","IT","ECE"),
Marks=c(85,90,78,88)
)
# Bar Chart
ggplot(data,aes(Department,Marks,fill=Department))+
geom_bar(stat="identity")
# Box Plot
ggplot(data,aes(x="",y=Marks))+
geom_boxplot(fill="yellow")
# Histogram
ggplot(data,aes(Marks))+
geom_histogram(fill="red",bins=5)
```





4. Scenario

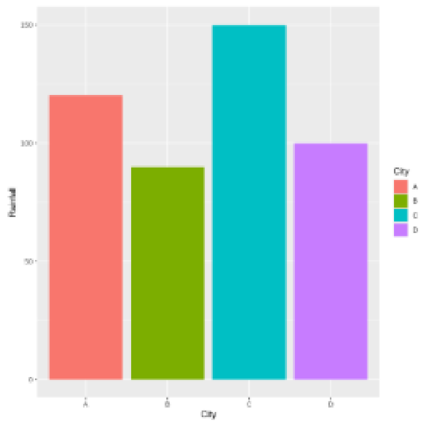
Climate and Rainfall Distribution Study

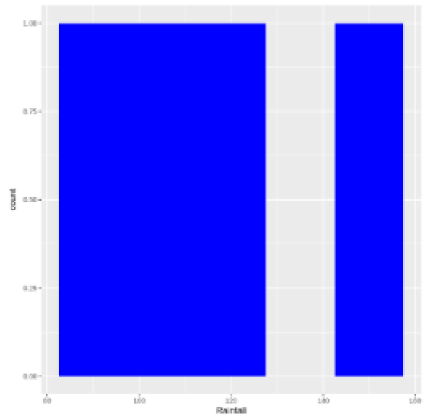
Student Task

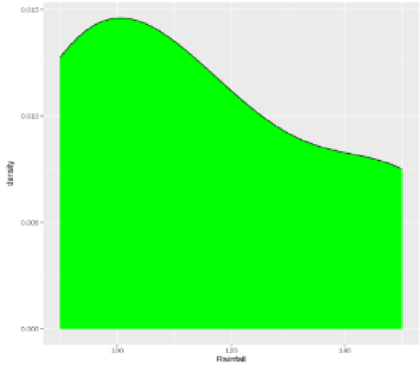
Meteorological data contains monthly rainfall, temperature, and humidity for multiple cities. Produce histograms, density plots, box plots, and bar charts. Evaluate which visualization best represents seasonal rainfall variation and climate distribution. Compare at least three visualizations and justify the most suitable one for environmental researchers.

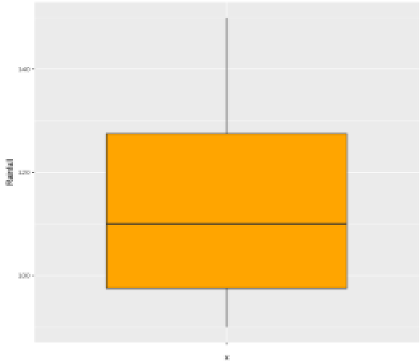
```
library(ggplot2)
data <- data.frame(
City=c("A","B","C","D"),
Rainfall=c(120,90,150,100)
)
# Bar Chart
ggplot(data,aes(City,Rainfall,fill=City))+
geom_bar(stat="identity")
```

```
# Histogram
ggplot(data,aes(Rainfall))+
geom_histogram(fill="blue",bins=5)
# Density Plot
ggplot(data,aes(Rainfall))+
geom_density(fill="green")
# Box Plot
ggplot(data,aes(x="",y=Rainfall))+
geom_boxplot(fill="orange")
```









5. Scenario

Bank Loan Risk Assessment

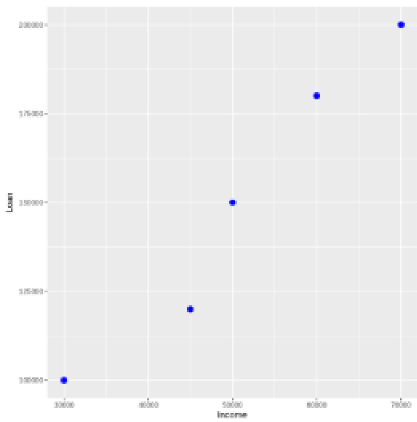
Student Task

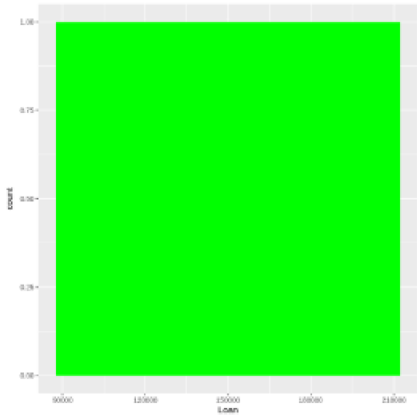
A financial institution has customer loan amounts, repayment history, income levels, and credit scores. Develop visualizations showing loan distributions and customer segments. Evaluate which visualization best supports loan approval decisions by highlighting risk patterns, skewness, and outliers. Recommend the visualization that should be included in a management dashboard.

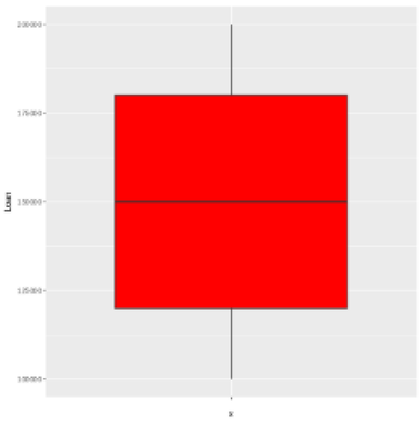
Code:

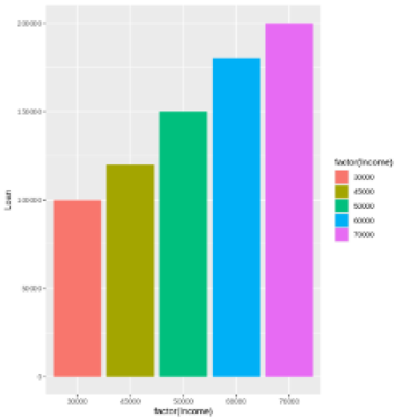
```
library(ggplot2)
data <- data.frame(
Income=c(30000,50000,45000,60000,70000),
Loan=c(100000,150000,120000,180000,200000)
)
# Scatter Plot
ggplot(data,aes(Income,Loan))+
geom_point(size=3,color="blue")
# Histogram
```

```
ggplot(data,aes(Loan))+
geom_histogram(fill="green",bins=5)
# Box Plot
ggplot(data,aes(x="",y=Loan))+
geom_boxplot(fill="red")
# Bar Chart
ggplot(data,aes(factor(Income),Loan,fill=factor(Income)))+
geom_bar(stat="identity")
```






200000
175000
150000
125000
100000
Loan
x


200000
150000
100000
50000
0
Loan
30000
45000
50000
60000
70000
factor(Income)
factor(Income)
30000
45000
50000
60000
70000